

Programme 1

Complex numbers 1

Learning outcomes

When you have completed this Programme you will be able to:

- ☒ Recognize j as standing for $\sqrt{-1}$ and be able to reduce powers of j to $\pm j$ or ± 1
- ☒ Recognize that all complex numbers are in the form (real part) + j (imaginary part)
- ☒ Add, subtract and multiply complex numbers
- ☒ Find the complex conjugate of a complex number
- ☒ Divide complex numbers
- ☒ State the conditions for the equality of two complex numbers
- ☐ Draw complex numbers and recognize the parallelogram law of addition
- ☐ Convert a complex number from Cartesian to polar form and vice versa
- ☐ Write a complex number in its exponential form
- ☐ Obtain the logarithm of a complex number

Introduction

1 Ideas and symbols

The numerals and their associated arithmetic symbols were devised so that we could make written calculations and records of quantities and measurements. When a grouping of symbols such as $\sqrt{-1}$ arises to which there is no corresponding quantity we must ask ourselves why? Why does the grouping of symbols occur if there is no quantity associated with it? Often, the only way to answer such a question is to accept the grouping of symbols and carry on manipulating with it to see if any new ideas are forthcoming that will help answer the question. This we shall now do.

By the way, because $\sqrt{-1}$ can have no quantity associated with it we call it an **imaginary** number. This is to distinguish it from those numbers to which we can associate quantities; these we call **real** numbers. This is a most unfortunate choice of words because it gives the idea that those numbers we call 'real' are somehow more concrete and proper than their ethereal and elusive cousins that we call 'imaginary'. In fact all numbers are constructs of the human mind and imagination; no one type is more concrete or proper than any other and the numerals we use to describe numbers of any type are just symbols that permit us to manipulate numbers and so communicate ideas of number from one mind to another. When we put those numbers to use as in counting or measuring it is easy to confuse the idea of number and the real world to which that number is applied.

In an earlier Programme it was stated that there are some quadratics that cannot be factorized into two linear factors. In fact all quadratics can be factorized into two linear factors but with the present state of your knowledge you cannot always find them. This will now be remedied.

Read on and see how

The symbol j

2 Quadratic equations

The solutions of the quadratic equation:

$$x^2 - 1 = 0$$

are:

$$x = \dots\dots\dots \text{ and } x = \dots\dots\dots$$

The answers are in the next frame

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$x = 1 \text{ and } x = -1$

Because:

$x^2 - 1 = 0$ can be written as $x^2 = 1$ by adding 1 to each side of the equation. The solution is then given as:

$$\begin{aligned} x &= 1^{\frac{1}{2}} = \pm\sqrt{1} \\ &= 1 \text{ or } -1 \end{aligned}$$



Try this next one.

The solutions to the quadratic equation:

$$x^2 + 1 = 0$$

are:

$x = \dots\dots\dots$ and $\dots\dots\dots$

Again, the answers are in the next frame

$$x = \sqrt{-1} \text{ and } x = -\sqrt{-1}$$

4

Because

$x^2 + 1 = 0$ can be written as $x^2 = -1$ by subtracting 1 from each side of the equation. The solution is then given as:

$$x = (-1)^{\frac{1}{2}} = \pm\sqrt{-1} = \sqrt{-1} \text{ or } -\sqrt{-1}$$

This notation is rather clumsy so we denote $\sqrt{-1}$ by the symbol j . This means that we can write the solution of the equation $x^2 + 1 = 0$ as:

$\dots\dots\dots$ and $\dots\dots\dots$

Next frame

$$x = j \text{ and } x = -j$$

5

Because

$$j = \sqrt{-1} \text{ and so } -j = -\sqrt{-1}$$

Let's take a closer look at quadratic equations in the next frame

We have already seen that the general quadratic equation is of the form:

$$ax^2 + bx + c = 0 \text{ with solution } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For example, if:

$2x^2 + 9x + 7 = 0$, then we have $a = 2$, $b = 9$, and $c = 7$ and so:

$$x = \frac{-9 \pm \sqrt{81 - 56}}{4} = \frac{-9 \pm \sqrt{25}}{4} = \frac{-9 \pm 5}{4}$$

$$\therefore x = -\frac{4}{4} \text{ or } -\frac{14}{4}$$

$$\therefore x = -1 \text{ or } -3.5$$

That was straightforward enough, but if we solve the equation $5x^2 - 6x + 5 = 0$ in the same way, we get

$$x = \frac{6 \pm \sqrt{36 - 100}}{10} = \frac{6 \pm \sqrt{-64}}{10}$$

and the next stage is now to determine the square root of -64 .

- Is it (a) 8
(b) -8
(c) neither?

Next frame

7

neither

It is, of course, neither, since +8 and -8 are the square roots of 64 and not of -64. In fact:

$$\sqrt{-64} = \sqrt{-1 \times 64} = \sqrt{-1} \times \sqrt{64} = j8 \text{ an imaginary number}$$

Similarly:

$$\sqrt{-36} = \sqrt{-1} \sqrt{36} = j6$$

$$\sqrt{-7} = \sqrt{-1} \sqrt{7} = j2.646$$

So $\sqrt{-25}$ can be written as

8

j5

We now have a way of finishing off the quadratic equation we started in Frame 6.

$$5x^2 - 6x + 5 = 0 \text{ therefore } x = \frac{6 \pm \sqrt{36 - 100}}{10} = \frac{6 \pm \sqrt{-64}}{10}$$

$$\therefore x = \frac{6 \pm j8}{10} \therefore x = 0.6 \pm j0.8$$

$$\therefore x = 0.6 + j0.8 \text{ or } x = 0.6 - j0.8$$

Just try one yourself to make sure that you have followed all this. The solution of the quadratic $x^2 + x + 1 = 0$ equation is

$x = \dots\dots\dots$ and $x = \dots\dots\dots$

Next frame

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$$x = \frac{-1 + j\sqrt{3}}{2} \text{ and } x = \frac{-1 - j\sqrt{3}}{2}$$

Because:

$$x^2 + x + 1 = 0 \text{ so that } a = 1, b = 1 \text{ and } c = 1$$

$$\text{Therefore } x = \frac{-1 \pm \sqrt{1 - 4}}{2} = \frac{-1 \pm j\sqrt{3}}{2},$$

$$\text{that is } x = \frac{-1 + j\sqrt{3}}{2} \text{ or } x = \frac{-1 - j\sqrt{3}}{2}$$

Now let's look a little closer at that number j .

Move to the next frame

Powers of j

Positive integer powers

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Because j represents $\sqrt{-1}$, it is clear that:

$$j^2 = -1$$

$$j^3 = (j^2)j = -j$$

$$j^4 = (j^2)^2 = (-1)^2 = 1$$

$$j^5 = (j^4)j = j$$

Note especially the third result: $j^4 = 1$. Every time a factor j^4 occurs, it can be replaced by the factor 1, so that j raised to a positive power is reduced to one of the four results $\pm j$ or ± 1 . For example:

$$j^9 = (j^4)^2 j = (1)^2 j = 1 \cdot j = j$$

$$j^{20} = (j^4)^5 = (1)^5 = 1$$

$$j^{30} = (j^4)^7 j^2 = (1)^7 (-1) = 1(-1) = -1$$

$$j^{15} = (j^4)^3 j^3 = (1)^3 (-j) = -j$$

So, in the same way:

(a) $j^{42} = \dots\dots\dots$

(b) $j^{12} = \dots\dots\dots$

(c) $j^{11} = \dots\dots\dots$

The answers are in the next frame

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$$(a) \ j^{42} = -1$$

$$(b) \ j^{12} = 1$$

$$(c) \ j^{11} = -j$$

Because:

$$j^{42} = (j^4)^{10} j^2 = (1)^{10} (-1) = -1$$

$$j^{12} = (j^4)^3 = (1)^3 = 1$$

$$j^{11} = (j^4)^2 j^3 = (1)^2 (-j) = -j$$

Now for negative powers in the next frame

Negative integer powers

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Negative powers follow from the reciprocal of j . Because

$$j^2 = -1$$

we see that by dividing both sides by j that $j = -\frac{1}{j} = -j^{-1}$ so that:

$$j^{-1} = -j$$

$$j^{-2} = (j^2)^{-1} = (-1)^{-1} = -1$$

$$j^{-3} = (j^{-2}) j^{-1} = (-1)(-j) = j$$

$$j^{-4} = (j^{-2})^2 = (-1)^2 = 1$$



Note again the last result. Every time a factor j^{-4} occurs, it can be replaced by the number 1, so that j raised to a negative power is reduced to one of the four results $\pm j$ or ± 1 . For example:

$$j^{-8} = (j^{-4})^2 = (1)^2 = 1$$

$$j^{-19} = (j^{-4})^4 j^{-3} = (1)^4(j) = j$$

$$j^{-30} = (j^{-4})^7 j^{-2} = (1)^7(-1) = -1$$

$$j^{-15} = (j^{-4})^3 j^{-3} = (1)^3(-j) = -j$$

So, in the same way:

(a) $j^{-32} = \dots\dots\dots$

(b) $j^{-13} = \dots\dots\dots$

(c) $j^{-23} = \dots\dots\dots$

The answers are in the next frame

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(a) $j^{-32} = 1$ (b) $j^{-13} = -j$ (c) $j^{-23} = j$
--

Because:

$$j^{-32} = (j^{-4})^8 = (1)^8 = 1$$

$$j^{-13} = (j^{-4})^3 j^{-1} = (1)^3(-j) = -j$$

$$j^{-23} = (j^{-4})^5 j^{-3} = (1)^5(j) = j$$

Complex numbers

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In Frame 8 we saw that the quadratic equation $5x^2 - 6x + 5 = 0$ had two solutions, one of which was $x = 0.6 + j0.8$. This consists of two separate terms, 0.6 and $j0.8$ which cannot be combined any further – it is a **mixture** of the real number 0.6 and the imaginary number $j0.8$. This mixture is called a **complex** number.

The word ‘complex’ means either:

- (a) complicated, or
- (b) mixture

Answer in the next frame

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mixture

A complex number is a mixture of a real number and an imaginary number. The symbol z is used to denote a complex number.

In the complex number $z = 3 + j5$ the number 3 is called the *real part* of z and is denoted as $\text{Re}(z)$. The number 5 is called the *imaginary part* of z and is denoted $\text{Im}(z)$.

So $\text{Re}(3 + j5) = 3$ and $\text{Im}(3 + j5) = 5$



Notice: the imaginary part of a complex number is a real number so that:

$$\text{Im}(3 + j5) = 5 \text{ and not } j5$$

So, a complex number $z = \text{Re}(z) + j\text{Im}(z)$.

In the complex number $z = 2 + j7$ the real and imaginary parts are

$$\text{Re}(z) = \dots\dots\dots \text{ and } \text{Im}(z) = \dots\dots\dots$$

Go to the next frame

$$\text{Re}(z) = 2 \text{ and } \text{Im}(z) = 7 \text{ (not } j7\text{)}$$

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Complex numbers have many applications in engineering and science. To use them, we must know how to carry out the usual arithmetical operations.

Addition and subtraction

This is easy, as a few examples will show:

$$(4 + j5) + (3 - j2)$$

Although the real and imaginary parts cannot be combined, we can remove the brackets and total up terms of the same kind:

$$\begin{aligned} (4 + j5) + (3 - j2) &= 4 + j5 + 3 - j2 = (4 + 3) + j(5 - 2) \\ &= 7 + j3 \end{aligned}$$

Another example:

$$\begin{aligned} (4 + j7) - (2 - j5) &= 4 + j7 - 2 + j5 = (4 - 2) + j(7 + 5) \\ &= 2 + j12 \end{aligned}$$

So in general, $(a + jb) + (c + jd) = (a + c) + j(b + d)$

Now you do this one:

$$(5 + j7) + (3 - j4) - (6 - j3) = \dots\dots\dots$$

$$2 + j6$$

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$$\begin{aligned} \text{since } (5 + j7) + (3 - j4) - (6 - j3) \\ &= 5 + j7 + 3 - j4 - 6 + j3 \\ &= (5 + 3 - 6) + j(7 - 4 + 3) = 2 + j6 \end{aligned}$$

Now you do these in the same way:

$$\begin{aligned} \text{(a) } (6 + j5) - (4 - j3) + (2 - j7) &= \dots\dots\dots \\ \text{and (b) } (3 + j5) - (5 - j4) - (-2 - j3) &= \dots\dots\dots \end{aligned}$$

$$\text{(a) } 4 + j \quad \text{(b) } j12$$

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Here is the working:

$$\begin{aligned} \text{(a) } (6 + j5) - (4 - j3) + (2 - j7) \\ &= 6 + j5 - 4 + j3 + 2 - j7 \\ &= (6 - 4 + 2) + j(5 + 3 - 7) = 4 + j \end{aligned}$$



$$\begin{aligned}
 \text{(b)} \quad & (3 + j5) - (5 - j4) - (-2 - j3) \\
 &= 3 + j5 - 5 + j4 + 2 + j3 \quad (\text{Take care with signs!}) \\
 &= (3 - 5 + 2) + j(5 + 4 + 3) \\
 &= 0 + j12 = j12
 \end{aligned}$$

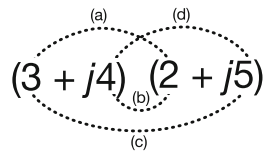
This is very easy then, so long as you remember that the real and the imaginary parts must be treated quite separately – just like x 's and y 's in an algebraic expression.

On to Frame 19

19 Multiplication

Take as an example: $(3 + j4)(2 + j5)$

These are multiplied together in just the same way as you would determine the product $(3x + 4y)(2x + 5y)$.



Form the product terms of

- (a) the two left-hand terms
- (b) the two inner terms
- (c) the two outer terms
- (d) the two right-hand terms

$$\begin{aligned}
 &= 6 + j8 + j15 + j^2 20 \\
 &= 6 + j23 - 20 \quad (\text{since } j^2 = -1) \\
 &= -14 + j23
 \end{aligned}$$

Likewise, $(4 - j5)(3 + j2) \dots\dots\dots$

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$22 - j7$

Because

$$\begin{aligned}
 (4 - j5)(3 + j2) &= 12 - j15 + j8 - j^2 10 \\
 &= 12 - j7 + 10 \quad (j^2 = -1) \\
 &= 22 - j7
 \end{aligned}$$

If the expression contains more than two factors, we multiply the factors together in stages:

$$\begin{aligned}
 (3 + j4)(2 - j5)(1 - j2) &= (6 + j8 - j15 - j^2 20)(1 - j2) \\
 &= (6 - j7 + 20)(1 - j2) \\
 &= (26 - j7)(1 - j2) \\
 &= \dots\dots\dots
 \end{aligned}$$

Finish it off

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$12 - j59$

Because

$$\begin{aligned}
 (26 - j7)(1 - j2) &= 26 - j7 - j52 + j^2 14 \\
 &= 26 - j59 - 14 = 12 - j59
 \end{aligned}$$

Note that when we are dealing with complex numbers, the result of our calculations is also, in general, a complex number.

Now you do this one on your own.

$$(5 + j8)(5 - j8) = \dots\dots\dots$$

89

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Here it is:

$$\begin{aligned}(5 + j8)(5 - j8) &= 25 + j40 - j40 - j^2 64 \\ &= 25 + 64 \\ &= 89\end{aligned}$$

In spite of what we said above, here we have a result containing no j term. The result is therefore entirely real.

This is rather an exceptional case. Look at the two complex numbers we have just multiplied together. Can you find anything special about them? If so, what is it?

When you have decided, move on to the next frame

They are identical except for the middle sign in the brackets, i.e. $(5 + j8)$ and $(5 - j8)$

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A pair of complex numbers like these are called *conjugate* complex numbers and *the product of two conjugate complex numbers is always entirely real*.

Look at it this way:

$$(a + b)(a - b) = a^2 - b^2 \quad \text{Difference of two squares}$$

Similarly

$$\begin{aligned}(5 + j8)(5 - j8) &= 5^2 - (j8)^2 = 5^2 - j^2 8^2 \\ &= 5^2 + 8^2 \quad \text{Sum of two squares} \\ &= 25 + 64 = 89\end{aligned}$$

Without actually working it out, will the product of $(7 - j6)$ and $(4 + j3)$ be

- (a) a real number
- (b) an imaginary number
- (c) a complex number?

A complex number

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since $(7 - j6)(4 + j3)$ is a product of two complex numbers which are *not* conjugate complex numbers or multiples of conjugates.

Remember: Conjugate complex numbers are identical except for the signs in the middle of the brackets.

	$(4 + j5)$ and $(4 - j5)$	<i>are</i>	conjugate complex numbers
	$(a + jb)$ and $(a - jb)$	<i>are</i>	conjugate complex numbers
but	$(6 + j2)$ and $(2 + j6)$	<i>are not</i>	conjugate complex numbers
	$(5 - j3)$ and $(-5 + j3)$	<i>are not</i>	conjugate complex numbers

So what must we multiply $(3 - j2)$ by, to produce a result that is entirely real?

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 $(3 + j2)$ or a multiple of it.

because the conjugate of $(3 - j2)$ is identical to it, except for the middle sign, i.e. $(3 + j2)$, and we know that the product of two *conjugate* complex numbers is always real.

Here are two examples:

$$(3 - j2)(3 + j2) = 3^2 - (j2)^2 = 9 - j^2 4 \\ = 9 + 4 = 13$$

$$(2 + j7)(2 - j7) = 2^2 - (j7)^2 = 4 - j^2 49 \\ = 4 + 49 = 53$$

... and so on.

Complex numbers of the form $(a + jb)$ and $(a - jb)$ are called complex numbers.

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conjugate

Now you should have no trouble with these:

(a) Write down the following products

$$\begin{array}{ll} \text{(i)} & (4 - j3)(4 + j3) \quad \text{(ii)} \quad (4 + j7)(4 - j7) \\ \text{(iii)} & (a + jb)(a - jb) \quad \text{(iv)} \quad (x - jy)(x + jy) \end{array}$$

(b) Multiply $(3 - j5)$ by a suitable factor to give a product that is entirely real.

When you have finished, move on to Frame 27

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Here are the results in detail:

$$\text{(a) (i)} \quad (4 - j3)(4 + j3) = 4^2 - j^2 3^2 = 16 + 9 = 25$$

$$\text{(ii)} \quad (4 + j7)(4 - j7) = 4^2 - j^2 7^2 = 16 + 49 = 65$$

$$\text{(iii)} \quad (a + jb)(a - jb) = a^2 - j^2 b^2 = \boxed{a^2 + b^2}$$

$$\text{(iv)} \quad (x - jy)(x + jy) = x^2 - j^2 y^2 = \boxed{x^2 + y^2}$$

(b) To obtain a real product, we can multiply $(3 - j5)$ by its conjugate, i.e. $(3 + j5)$, giving:

$$(3 - j5)(3 + j5) = 3^2 - j^2 5^2 = 9 + 25 = 34$$

Now move on to the next frame

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Division

Now let us deal with division.

Division of a complex number by a real number is easy enough:

$$\frac{5 - j4}{3} = \frac{5}{3} - j\frac{4}{3} = 1.67 - j1.33 \text{ (to 2 dp)}$$

But how do we manage with $\frac{7 - j4}{4 + j3}$?



If we could, somehow, convert the denominator into a real number, we could divide out as in the example above. So our problem is really, how can we convert $(4 + j3)$ into a completely real denominator – and this is where our last piece of work comes in.

We know that we can convert $(4 + j3)$ into a completely real number by multiplying it by its

conjugate

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i.e. the same complex number but with the opposite sign in the middle, in this case $(4 - j3)$.

But if we multiply the denominator by $(4 - j3)$, we must also multiply the numerator by the same factor:

$$\frac{7 - j4}{4 + j3} = \frac{(7 - j4)(4 - j3)}{(4 + j3)(4 - j3)} = \frac{28 - j37 - 12}{16 + 9} = \frac{16 - j37}{25}$$

$$\frac{16}{25} - j\frac{37}{25} = 0.64 - j1.48$$

and the job is done. To divide one complex number by another, therefore, we multiply numerator and denominator by the conjugate of the denominator. This will convert the denominator into a real number and the final step can then be completed.

Thus, to simplify $\frac{4 - j5}{1 + j2}$, we shall multiply top and bottom by

the conjugate of the denominator, i.e. $(1 - j2)$

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If we do that, we get:

$$\frac{4 - j5}{1 + j2} = \frac{(4 - j5)(1 - j2)}{(1 + j2)(1 - j2)} = \frac{4 - j13 - 10}{1 + 4}$$

$$= \frac{-6 - j13}{5} = \frac{-6}{5} - j\frac{13}{5} = -1.2 - j2.6$$

Now here is one for you to do: Simplify $\frac{3 + j2}{1 - j3}$

When you have done it, move on to the next frame

 $-0.3 + j1.1$

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Because

$$\frac{3 + j2}{1 - j3} = \frac{(3 + j2)(1 + j3)}{(1 - j3)(1 + j3)} = \frac{3 + j11 - 6}{1 + 9} = \frac{-3 + j11}{10} = -0.3 + j1.1$$

Now do these in the same way:

(a) $\frac{4 - j5}{2 - j}$ (b) $\frac{3 + j5}{5 - j3}$ (c) $\frac{(2 + j3)(1 - j2)}{3 + j4}$

When you have worked these, move on to Frame 32 to check your results

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Here are the solutions in detail:

$$(a) \frac{4-j5}{2-j} = \frac{(4-j5)(2+j)}{(2-j)(2+j)} = \frac{8-j6+5}{4+1} = \frac{13-j6}{5} = \boxed{2.6-j1.2}$$

$$(b) \frac{3+j5}{5-j3} = \frac{(3+j5)(5+j3)}{(5-j3)(5+j3)} = \frac{15+j34-15}{25+9} = \frac{j34}{34} \neq$$

$$(c) \frac{(2+j3)(1-j2)}{(3+j4)} = \frac{2-j+6}{3+j4} = \frac{8-j}{3+j4}$$

$$= \frac{(8-j)(3-j4)}{(3+j4)(3-j4)}$$

$$= \frac{24-j35-4}{9+16} = \frac{20-j35}{25}$$

$$= \boxed{0.8-j1.4}$$

And now you know how to add, subtract, multiply and divide complex numbers

33**Equal complex numbers**

Now let us see what we can find out about two complex numbers which we are told are equal.

Let the numbers be $a + jb$ and $c + jd$

Then we have $a + jb = c + jd$

Rearranging terms, we get $a - c = j(d - b)$

In this last statement, the quantity on the left-hand side is entirely real, while that on the right-hand side is entirely imaginary, i.e. a real quantity equals an imaginary quantity! This seems contradictory and in general it just cannot be true. But there is *one* special case for which the statement can be true. That is when

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each side is zero

$$a - c = j(d - b)$$

can be true only if $a - c = 0$, i.e. $a = c$

and if $d - b = 0$, i.e. $b = d$

So we get this important result:

If two complex numbers are equal

(a) the two real parts are equal

(b) the two imaginary parts are equal

For example, if $x + jy = 5 + j4$, then we know $x = 5$ and $y = 4$

and if $a + jb = 6 - j3$, then $a = \dots\dots\dots$ and $b = \dots\dots\dots$

$$a = 6 \text{ and } b = -3$$

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Be careful to include the sign!

Now what about this one?

If $(a + b) + j(a - b) = 7 + j2$, find the values of a and b .

Well now, following our rule about two equal complex numbers, what can we say about $(a + b)$ and $(a - b)$?

$$a + b = 7 \text{ and } a - b = 2$$

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since the two real parts are equal and the two imaginary parts are equal.

This gives you two simultaneous equations, from which you can determine the values of a and b .

So what are they?

$$a = 4.5; b = 2.5$$

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$$\text{For } \begin{cases} a + b = 7 \\ a - b = 2 \end{cases} \begin{cases} 2a = 9 \\ 2b = 5 \end{cases} \therefore \begin{cases} a = 4.5 \\ b = 2.5 \end{cases}$$

We see then that an equation involving complex numbers leads to a pair of simultaneous equations by putting

- (a) the two real parts equal
- (b) the two imaginary parts equal

This is quite an important point to remember.

Review exercise



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- 1 Simplify (a) j^{12} (b) j^{10} (c) j^{-7}
- 2 Simplify:
 - (a) $(5 - j9) - (2 - j6) + (3 - j4)$
 - (b) $(6 - j3)(2 + j5)(6 - j2)$
 - (c) $(4 - j3)^2$
 - (d) $(5 - j4)(5 + j4)$
- 3 Multiply $(4 - j3)$ by an appropriate factor to give a product that is entirely real. What is the result?
- 4 Simplify $\frac{3 - j2}{1 + j}$ into the form $a + jb$
- 5 If $\frac{2 - a - j5}{1 - jb} = 3 - j5$ find a and b

When you have completed this exercise, move on to Frame 39